Customer Stories / Games / United States

2024

# Helping Players Truly Own In-Game Assets Using AWS with Mystic Moose and Sequence

Learn how Mystic Moose and Sequence built a Web3 gaming world that integrates blockchain with cloud technology on AWS.

Overview | Opportunity | Solution | Outcome | AWS Services Used

**Scales**

to nearly 100,000 players and countin

**Faster**

**Minted**



**Facilitates**

true ownership of digital assets

## Overview

Video game enthusiasts are constantly searching for unique experiences that stand out from all the rest. These games need to be immersive with fast performance and low latency to provide the best experience possible. Mystic Moose, with its debut title, Mojo Melee, and its Planet Mojo community, is defining the next generation of gaming on Amazon Web Services (AWS). This is powered by Sequence, an all-in-one development platform for building Web3 games.

To create Mojo Melee, the studio adopted AWS services and worked with Sequence to combine blockchain with online gaming. Mojo Melee provides a unique, immersive experience for its players, who enjoy enhanced gameplay and true digital ownership of their in-game assets.


## Opportunity | Using AWS and Sequence to Create a New Gaming Experience for Mystic Moose

Founded in 2021 by game industry veterans, Mystic Moose is an independent game studio that blends online gaming with blockchain and Web3 technologies. It wants to improve player engagement by integrating unique features into the games that it creates, such as non-fungible tokens (NFTs).

Planet Mojo is a network of interoperable games that are built by Mystic Moose and set inside an alien planet with an evolving narrative. Mojo Melee, an auto chess battler, is the studio's first title. To bring this immersive, user-driven experience to life, Mystic Moose needed to choose the right technology to power its game.

"Given the online nature of the game, ensuring robust scalability and performance was extremely crucial," says Ted Newell, director of marketing at Mystic Moose. "This involved optimizing the game's infrastructure for smooth gameplay across different devices and systems. We also needed to manage the backend to handle a growing number of players without compromising on performance."

Mystic Moose chose AWS and Sequence to turn its vision into a reality. Due to experience usin es, its team was confident that AWS could meet the game's requirements for performance and scale with Sequence, the studio could seamlessly integrate the blockchain capabilities that the game req

"With Sequence, you can quickly incorporate Web3 experiences into your game without sacrificing your players' experience," says Megan Doyle, growth marketing lead at Sequence. "From collectible cosmetics and ownable rewards to fully on-chain experiences, Sequence's modular stack solves the technical challenges so studios can focus on creative tasks."

## Solution | Building an Immersive Gaming World with Nearly 100,000 Players and Counting

To create Mojo Melee, Mystic Moose first built the game's core infrastructure on AWS, adopting services such as Amazon Simple Storage Service (Amazon S3), an object storage service built to retrieve any amount of data from anywhere. The studio uses Amazon S3 as scalable storage for all game assets, facilitating the fast storage and retrieval of data such as graphics, animations, and player profiles. Amazon CloudFront—a service that helps securely deliver content with low latency and high transfer speeds—serves as the content delivery network. Mystic Moose uses this service to cache game assets and quickly deliver these assets to players from the locations closest to them.

"One version of Mojo Melee is played in a web browser, and hosting such a large, complex game cost effectively would have been hard to imagine without Amazon S3," says Michael Levine, founder of Mystic Moose. "Amazon CloudFront not only caches static content, but it also improves download performance worldwide for our players."

On AWS, nearly 100,000 players and counting can quickly load Mojo Melee from anywhere in the world. Using AWS Lambda, a serverless, event-driven compute service, Mystic Moose can scale the game naturally as it grows in popularity and elastically as demand spikes—without any manual intervention. The studio also paired Amazon DynamoDB, a serverless, NoSQL, fully managed database, with serverless API calls for metadata management. This efficient setup helped the studio manage traffic bursts during special in-game events while keeping costs low.

Additionally, Mystic Moose uses Amazon S3 and Amazon DynamoDB to manage NFTs. Amazon S3 is used for storing and updating new image files whenever NFTs are minted or avatars are altered. These changes are immediately visible to users online. Concurrently, it uses Amazon DynamoDB to manage evolving NFT metadata as players earn awards and level up in the game. Mystic Moose is also evaluating other AWS services tailored for blockchain builders, such as Amazon Managed Blockchain (AMB) Access Polygon, which provides serverless access to Polygon's standard blockchain APIs.



Mystic Moose uses APIs from Sequence to create Web3 wallets and mint players' NFTs; Polygo
blockchain protocol where these NFTs are deployed. It adopted the Sequence Marketplace Engine to build its white-

label collectibles marketplace, which lets users purchase NFTs. Through social and email logins, players can establish a non-custodial, multichain smart wallet, where they can securely manage their digital assets and transactions within the game. To date, Mystic Moose has minted tens of thousands of NFTs.

“With AWS services and Sequence, we’ve created a Web3 world that’s not just about using blockchain for its own sake, but genuinely enhances the gaming experience,” says Levine. “We can push forward the idea that people can truly own their own digital assets and not be beholden to a centralized economy.”

## Outcome | Pushing the Boundaries of Online Gaming Using AWS and Blockchain Technologies

As of January 2024, Mojo Melee is available to play on desktop browsers and a newly released mobile app. In the future, Mystic Moose will continue to expand its game universe and experiment with Web3 and blockchain technologies to push the boundaries of online gaming.

Mojo Melee players are not just passive participants but active owners and traders in the game’s universe, with real-world value attributed to their in-game activities and assets. And using AWS services, they enjoy powerful, fast performance anywhere in the world.

“On AWS, players can load our game quickly, from anywhere on the planet, and get right into playing,” says Levine. “We needed to have fast load times for players, and AWS helped with this tremendously.”

## About Mystic Moose

Mystic Moose is an independent game studio and publisher based in Boston, Massachusetts. Founded in 2021, it focuses on blending traditional Web3 gaming with the latest blockchain technology.

## AWS Services Used

### Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

Learn more »

### Amazon CloudFront

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Amazon CloudFront is a content delivery network (CDN) service built for high performance, security, and developer convenience.

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## AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

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## Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL database service that enables you to develop modern applications at any scale.

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